Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/23/2023
PI Name:	Lelkes, Peter Istvan Ph.D.		
Project Title:	The Effects of Aggregate Culture in Modeled Microgravity and Partial Gravity on Endocrine Differentiation of Induced Pluripotent Stem Cells (iPSC)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	pilelkes@temple.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-204-3307
Organization Name:	Temple University		
PI Address 1:	Department of Bioengineering		
PI Address 2:	1947 N 12th St, Room 811		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19122-6018	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
Contact Email:	annamaria.j.ruby@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/26/2022 per NSSC information (Ed., 8/5/21) NOTE: End date changed to 7/26/2021 per NSSC information (Ed., 6/3/20)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):			
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Task Description:	The long-term goal of this proposal is to study the effects of modeled microgravity and partial gravity on aggregate (organoid) cultures of induced pluripotent stem cells (iPSCs). Our working hypothesis, based on some preliminary data, is that hypogravity, i.e., modeled microgravity and/or partial gravity, will enhance iPSC differentiation into the endodermal lineages. Specifically, we will compare the efficiency of two distinct microgravity gravity / partial gravity simulating devices, i.e., the Rotating Wall Vessel (RWV) bioreactor and Random Positioning Machine (RPM) to modulate the maintenance of iPSCs stemness and the directed differentiation of iPSC into definitive endoderm. In testing our hypothesis, we developed the following specific aims: Specific Aim 1. To characterize the differentiation of iPSC into definitive endoderm in 2D monolayer culture in normal, and simulated partial- (Moon, Mars) and microgravity; and in 3D aggregate suspension culture in partial- (RPM) and microgravity (RPM and RWV). Specific Aim 2. Examine the underlying genetic, epigenetic, molecular, and biochemical mechanisms of stem cell differentiation and behavior that are affected by aggregate culture, hypoxia, and hypogravity; and identify specific molecular and signaling factors important in modulation of differentiation. In these studies, we will use biochemical, immunological, and molecular biological approaches to analyze the differential expression of a set of lineage-specific specific marker genes and gene products under altered gravity conditions. Subsequently, we will identify specific genetic, epigenetic, and molecular mechanisms that underlie and are influenced by partial- and microgravity and compare these biologic factors with those from induced iPSC differentiation, in both aggregate and monolayer culture, in standard gravity. The latter data will be analyzed by systems- / bioinformatics-based analytical approaches, such as principle component analysis and network analysis, which will help to identify specific hypogravity-sensitive cellular signaling molecules and pathways that may be differentially affected by varying levels of partial/microgravity, and which could be further explored in actual space experiments. In addition to having available both hypogravity simulating devices, we have introduced several technical innovations, such as modifications in the RWV bioreactors, enhanced oxygenation capabilities in RPM culture, and a high-resolution optical monitoring system that allows real-time visualization of intracellular signaling events. We anticipate that we will demonstrate a superior environment for iPSC differentiation in simulated and/or partial gravity conditions and that knowledge gained from these studies will contribute to technological and medical advancements both on Earth and in space and enhance our understanding of the adaptation and function of biological processes in altered gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Stem cell behavior and differentiation in simulated hypogravity mimicking partial gravity (PG) in extraterrestrial planets (Mars, Moon) and also simulated microgravity conditions modeling microgravity conditions, in LEO and the specific advantages of three-dimensional (3D) cell culture (compared to traditional 2D monolayer culture), represents an important, under-examined field of inquiry. First, cells in 3D aggregates (organoids, embryoid bodies, EBs) have been shown to better emulate in vivo biology and behavior than those in traditional 2D monolayer culture, and the examination of these systems may contribute to technical and medical advancements on earth. Second, a better understanding of stem cell behavior and differentiation in microgravity should have a considerable impact on NASA's Space Biology initiative, in terms of the understanding, adaptation, and potential amelioration of detrimental effects of microgravity. Finally, the development of 3D hiPSC culture systems in SMG and PG provides a relevant clinical platform to examine the effects of space travel on human physiology. We anticipate that we will demonstrate a superior environment for iPSC differentiation in simulated and/or partial gravity conditions and that knowledge gained from these studies will contribute to technological and medical advancements both on Earth and in space and enhance our understanding of the adaptation and function of biological processes in altered gravity.
Task Progress:	Our long-term goal was to study the effects of modeled microgravity and partial gravity on aggregate (organoid) cultures of induced pluripotent stem cells (iPSCs). Our working hypothesis, based on some preliminary data, was that hypogravity, i.e., simulated microgravity (SMG) and/or partial gravity (PG), will enhance iPSC differentiation into the endodermal lineages. Specifically, we proposed to compare the efficiency of two distinct microgravity gravity / partial gravity simulating devices, i.e., the rotating wall vessel (RWV) bioreactor and random positioning machine (RPM) to modulate the maintenance of iPSCs stemness and the directed differentiation of iPSC into definitive endoderm. In testing our hypothesis, we developed the following specific aims: Specific Aim 1. To characterize the differentiation of iPSC into definitive endoderm in 2D monolayer culture in normal, and simulated partial- (Moon, Mars), and microgravity; and in 3D aggregate suspension culture in partial- (RPM) and microgravity (RPM and RWV). Specific Aim 2. Examine the underlying genetic, epigenetic, molecular, and biochemical mechanisms of stem cell differentiation and behavior that are affected by aggregate culture, hypoxia, and hypogravity; and identify specific molecular and signaling factors important in the modulation of differentiation. We proposed using biochemical, immunological, and molecular biological approaches to analyze the differential expression of a set of lineage-specific marker genes and gene products under altered gravity conditions. We anticipated demonstrating a superior environment for iPSC differentiation in simulated and/or partial gravity conditions and that knowledge gained from these studies might contribute to technological and medical advancements both on Earth and in space and enhance our understanding of the adaptation and function of biological processes in altered gravity. We had to change our plans and research strategies several times, first because of technical difficulties with the human iPSC cell in the first year of the grant. Hence, while we were trying to overcome these obstacles, we concomitantly carried out some of the proposed studies using mouse embryonic stem cells (mESCs) instead of human induced pluripotent stem cells (hiPSC). This approach was successful, and we concluded this part of our studies with meaningful results and a peer-reviewed publication (Oss-Ronen et al., 2020), and several conference presentations. By the beginning of 2020, we were back on track with using the human iPSC, when the COVID pandemic hit (March 2020). We were forced to shut down the laboratory for several months. By the fall of 2020, our labs were slowly reopened with lots of restrictions and severe disruptions of the supply chain for several additional months, leading to a tremendous downturn in our productivity. In addition, towards the end of the reporting period (Winter 2021/2022), the graduate student working on the project experienced severe COVID-related medical issues and left the project, impacting the experiments and the analysis of results. Hence, we could complete only a fraction of the aims and goals of our original proposal.

	In the first part of our study, while we were troubleshooting the problems of working with human iPSC, we investigated whether the 3D culture of mESCs in simulated microgravity (SMG) bioreactors (BRs) would be advantageous for directed DE (definitive endoderm) differentiation. To test our hypothesis, we examined several characteristic parameters, such as the morphology of the cells/aggregates and the expression of several markers of pluripotency and differentiation. The six different conditions in our protocol provided a system that compared the 2D versus 3D culture of the cells in a) maintenance medium (MM) versus an established differentiation medium (DM), and b) under static versus dynamic conditions in the SMG BRs. Our results indicated that the cells preferentially lost pluripotency and showed a more significant increase in the expression of DE markers in DM in the rotary BR environment compared to static EBs. The goal of the second part of our study was to explore the effect of hypogravity / partial gravity (gravity conditions on Moon and Mars) on the pluripotency (stemness) and directed DE differentiation of cultured hiPSC in 2D and 3D cultures, as originally proposed. To simulate reduced gravity / partial gravity (PG), all experiments were carried out in the second generation of the software-driven Random Positioning Machine. There are several major conclusions from our studies: 1. Our data so far support our hypothesis that exposure to hypogravity and 3D culture downregulates pluripotency in both mESC and hiPSC. 2. To facilitate comparison between different experiments, we introduced a dimensionless "Differentiation Index, DI", in analogy to the macrophage polarization index. 3. Based on the magnitude of the DI, we can define a threshold signature of the dose-dependent impairment of hiPSC differentiation: DIEarth~ DIMars >> DIMoon~ DIMicro. 4. Directed DE differentiation is enhanced in 3D aggregates of mESC, but generally downregulated in hiPSC, irrespective of the culture conditions. This finding, observed in hiPSC, but not in mESC, contradicts our original working hypothesis. Going forward, this new premise will have to be independently validated. If it holds, the new premise will require rethinking and reformulating our hypothesis. 5. Finally, while cells and organoids in both the RWV BR and in the RPM are exposed to hypogravity and experience very low shear and turbulence, the differences between the environment in the ground-based simulating systems and true space conditions are not negligible. Hence, we caution against uncritical extrapolation from simulated SMG and PG experiments on Earth to predict the effects of microgravity in space. Further testing under low-Earth orbit (LEO) spaceflight conditions on board the International Space Station (ISS) and subsequently on the Moon or Mars will be required to unequivocally assess the effects of actual extraterrestrial hypogravity on DE differentiation.

Bibliography Type:	Description: (Last Updated: 09/28/2023)
Articles in Peer-reviewed Journals	Phelan M, Kruczek K, Wilson J, Brooks M, Drinnan C, Regent F, Gerstenhaber JA, Lelkes PI, Swaroop A, Li T. "Soy protein nanofiber scaffolds for uniform maturation of hiPSC-derived retinal pigment epithelium." Tissue Eng Part C Methods. 2020 Aug;26(8):433-446. https://doi.org/10.1089/ten.tec.2020.0072 ; PMID: 32635833; PMCID: PMC7462036 , Aug-2020
Articles in Peer-reviewed Journals	Braveboy-Wagner J, Sharoni Y, Lelkes PI. "Nutraceuticals synergistically promote osteogenesis in cultured 7F2 osteoblasts and mitigate inhibition of differentiation and maturation in simulated microgravity." Int J Mol Sci. 2022 Jan;23(1):136. https://doi.org/10.3390/ijms23010136 ; PMID: 35008559; PMCID: PMC8745420 , Jan-2022
Articles in Peer-reviewed Journals	Braveboy-Wagner J, Lelkes PI. "Impairment of 7F2 osteoblast function by simulated partial gravity in a random positioning machine." npj Microgravity. 2022 Jun 7;8(1):20. https://doi.org/10.1038/s41526-022-00202-x ; PMID: 35672327; PMCID: PMC9174291 , Jun-2022
Articles in Peer-reviewed Journals	Licata JP, Schwab KH, Har-El YE, Gerstenhaber JA, Lelkes PI. "Bioreactor technologies for enhanced organoid culture." Int J Mol Sci. 2023 Jul 13;24(14):11427. Review. https://doi.org/10.3390/ijms241411427 ; PMID: 37511186; PMCID: PMC10380004 , Jul-2023